

How Much Energy Electrons Can Emit? Radiation Theory Based on the Lorentz-Dirac Equation and other Modified Equations

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The Lorentz-Dirac (LD) equation or its modified equations describe the motion of a classical point charge. We investigate the electron radiation in two classical examples which predict new results and can assess the validity of these equations. The first example is that, the radiation spectrum profiles from a 1-dimensional harmonic bound electron is different based on different equations. Second example demonstrates the synchrotron radiation from a ultra relativistic electron in a strong magnetic field, which shows very different behavior compared to a classical one. For the original LD equation we show that the power emission which depends on the initial acceleration condition significantly decreases for $B\gamma > 10^{16}$ G compared to the classical result $P_{\text{em}} \propto v^2\gamma^2$. The total emitted energy will be higher than the kinetic energy change due to the radiation damping effects. For the modified LD equation, we derive the exact analytic solutions for electron motion and radiation, and show the similarities and differences with the results derived from the LD equation, which can be expected to be detected in principle and applied in magnetic neutron stars and magnetars. We also discuss the electron radiation using the Eliezer's equation.

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I. INTRODUCTION

The reaction of a classical charged particle to its own radiation has been studied for one century since Lorentz and Abraham [1, 2]. Since the radiation carries away energy and momentum, there must be a radiation damping force or say the self-force on the charged particle to influence the motion of the charge. The most prominent equation of motion for a charged particle in the context of classical relativistic electrodynamics is the Lorentz-Dirac (LD) equation [2, 3]. Generally speaking, the LD equation is derived by subtracting out the infinite self-energy of the particle in a Lorentz invariant way, and is consistent with the energy-momentum conservation laws in the Minkowski space.

However, the LD equation still meets difficulties because it unusually involves the derivative of the acceleration vector. The well-known problems of the LD equation are the runaway, preacceleration and ununiqueness solutions (see Poisson [4] for a review). Dirac [3] argued that some of the weird behaviors may be eliminated if we consider the electron particle is supposed to have a finite size rather than a singular point. Since then, several alternative equations of motions have been proposed [4–11]. However, in classical electrodynamics which equation is the correct one is still under debate. Besides the discussion on the motion of equation itself, some other works have extended the LD results into curved spacetime background [12, 13], and into the framework of quantum electrodynamics, e.g., [14–17].

The theoretical LD equation and its modified equations have their application in many aspects, such as in linear acceleration process [18, 19], laser fields environment [20, 21], central force or the Rutherford scattering problem [22–25], Compton scattering process [26], and the environment of very strong magnetic fields with ultrarelativistic electrons [27–30]. At the same time, people try to study the results from these applications and go back to assess the validity of the motion of equations for a classical charged particle.

In this paper, we focus on two classical problems, the classical 1-dimensional harmonically bound electron oscillator and the synchrotron radiation of ultrarelativistic electrons in very strong magnetic fields. Although both of these two problems have been studied by some other authors [27–31], we still discover new results from these “textbook” problems. We compare the solutions and related spectrum profile of the harmonic oscillator based on the original LD equation and one of its modified form showed by [4] and [11], and provide the possible way in principle to assess the correctness of these equations. Furthermore, we demonstrate the results of synchrotron radiation with energy and power emission, which can be significantly different with the classical results. The previous study on the radiation reaction in strong fields focus on the electron motion itself and always consider the radiation damping with a lowest order of correction. We mainly focus on the radiation process, and we also try to find the solution in another limit case $B\gamma > 10^{16}$ G, where the perturbation method which take the classical result as the zeroth-order solution is invalid in this case. Actually we can find the exact analytic solution of electron motion perpendicular to

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the magnetic fields as well as the synchrotron emission based on the modified LD equation mentioned before. We shows the significant difference of radiation process between the classical estimate and that includes the radiation damping reaction. Moreover, we shows the difference of energy and power radiation in strong fields based on different motion of equations.

This paper is organized as follows. In Section II we study the motion of a particle which is harmonically bound to a center of force proportional to the distance, and show the spectrum profile of the radiation power. In Section III we study the synchrotron radiation in a very strong field based on the LD equation. We use the perturbation method for both of the two limits $B\gamma \ll 10^{16}$ G and $B\gamma \gg 10^{16}$ G. According to the LD equation, the energy released should be greater than the electron kinetic change γmc^2 , but the exact energy emission for $B\gamma \gg 10^{16}$ depend on the initial acceleration condition as well as the velocity condition (i.e., the Lorentz factor). Next we calculate the corresponding radiation results in Section IV based on the modified LD equation. We first demonstrate the different spectrum profile for the harmonic oscillator in Section IV A, and show the exact analytic solution of the electron motion with its radiation power in Section IV B. To our knowledge it is the first time to show the significant difference between radiation with the classical calculation $P_{\text{em}} \propto \beta^2 \gamma^2$ and with radiation reaction which $P_{\text{em}} \sim \text{Const.}$ for $B\gamma \gg 10^{16}$ G. This conclusion can be applied on the surface or inside the magnetic neutron stars and magnetars in astrophysics. Furthermore, we calculate the energy radiation in strong field based on the Eliezer's equation with a parameter k as an supplementary of the synchrotron radiation theory in this paper.

II. HARMONICALLY BOUND CHARGES

For a particle which is harmonically bound to a center of force $\mathbf{F} \propto \mathbf{r}$, there is always a small damping of the oscillations by the radiation reaction force, which can be described by a third-order differential equation in the Newtonian case [32], or derived using the 1-dimensional LD equation under the relativistic covariance. The space component of the equation is

$$\ddot{\xi} - b\xi^{(3)} + \frac{b}{c^2}\dot{\xi}(\ddot{\xi}^2 - c^2\dot{t}^2) = -\gamma\omega_0^2\xi, \quad (1)$$

where ξ is the displacement, $b = 2e^2/(3mc^3)$ is the damping parameter, $\omega_0^2 = \sqrt{k/m}$ is the characteristic frequency, $\dot{\xi} = d\xi/d\tau$ and $\ddot{\xi} = d^2\xi/d\tau^2$ are the four-velocity and four-acceleration of the oscillated charge with τ as the proper time. If we define $\tilde{\beta} = c^{-1}d\xi/d\tau$ to be different with $\beta = c^{-1}d\xi/dt$, thus we have $\gamma = (1 + \tilde{\beta}^2)^{1/2}$, $\dot{t} = \gamma$, $\ddot{t} = \dot{\gamma} = \tilde{\beta}\dot{\tilde{\beta}}/\gamma$, and $\ddot{\xi}^2 - c^2\dot{t}^2 = \ddot{\xi}^2(1 - \tilde{\beta}^2/\gamma^2) = \ddot{\xi}^2/\gamma^2$. The original equation (1) can be written as

$$\ddot{\xi} - b\xi^{(3)} + \frac{b}{c^2}\frac{\dot{\xi}\ddot{\xi}^2}{\gamma^2} = -\gamma\omega_0^2\xi, \quad (2)$$

We take the solution as the function of proper time τ as

$$\xi = A_0 \exp(-\lambda\tau) \cos \omega\tau. \quad (3)$$

In the non-relativistic case $\tilde{\beta} < 1$ we can expand the Lorentz factor as $\gamma = 1 + \frac{1}{2}\tilde{\beta}^2 - \frac{1}{8}\tilde{\beta}^4 + \dots$, and $\gamma^{-2} = 1 - \tilde{\beta}^2 + \tilde{\beta}^4 + \dots$. If we assume the damping factor $-\lambda\tau$ as a small number corresponding to the damping force $\mathcal{O}(b)$, we derive the zeroth-order equation as

$$-\omega^2 = -\left(1 + \frac{1}{4}\frac{A_0^2\omega^2}{c^2}e^{-2\lambda\tau} - \frac{3}{64}\frac{A_0^4\omega^4}{c^4}e^{-4\lambda\tau}\right)\omega_0^2, \quad (4)$$

where we expand γ and γ^{-2} to the order of $\mathcal{O}(\tilde{\beta}^4)$. Then we can derive the solution of the characteristic frequency ω as

$$\frac{\omega^2}{\omega_0^2} = \left[\left(1 - \frac{1}{2}\tilde{\beta}_0^2 e^{-2\lambda\tau} + \frac{1}{4}\tilde{\beta}_0^4 e^{-4\lambda\tau}\right)^{1/2} - \left(1 - \frac{1}{4}\tilde{\beta}_0^2 e^{-2\lambda\tau}\right) \right] \left(\frac{3}{32}\tilde{\beta}_0^4 e^{-4\lambda\tau} \right)^{-1}, \quad (5)$$

The approximated expression for the solution (5) expanding to the first order is

$$\frac{\omega^2}{\omega_0^2} = \left(1 - \frac{1}{4}\tilde{\beta}_0^2 e^{-2\lambda\tau}\right)^{-1}, \quad (6)$$

where we assume $\frac{3}{16}\tilde{\beta}_0^4 e^{-4\lambda\tau} \ll \left(1 - \frac{1}{4}\tilde{\beta}_0^2 e^{-2\lambda\tau}\right)^2$ and expand the term of square root in the numerator of solution (5) to the first order.

The first-order equation obtained from equation (2) reads

$$2\lambda - b\omega^2 \left(1 + \tilde{\beta}_0^2 \frac{\omega^2}{\omega_0^2} e^{-2\lambda\tau} \cos^2 \omega\tau - \tilde{\beta}_0^4 \frac{\omega^4}{\omega_0^4} e^{-4\lambda\tau} \sin^2 \omega\tau \cos^2 \omega\tau\right) \quad (7)$$

$$= -\lambda \left(\tilde{\beta}_0^2 e^{-2\lambda\tau} \cos^2 \omega\tau - \frac{1}{2}\tilde{\beta}_0^4 e^{-4\lambda\tau} \sin^2 \omega\tau \cos^2 \omega\tau\right), \quad (8)$$

or we have the solution

$$\lambda = \frac{1}{2}b\omega^2 \left(1 + \frac{\tilde{\beta}_0^2}{2} \frac{\omega^2}{\omega_0^2} e^{-2\lambda\tau} - \frac{\tilde{\beta}_0^4}{8} \frac{\omega^4}{\omega_0^4} e^{-4\lambda\tau}\right) \left(1 + \frac{\tilde{\beta}_0^2}{4} \frac{\omega^2}{\omega_0^2} e^{-2\lambda\tau} - \frac{\tilde{\beta}_0^4}{32} \frac{\omega^4}{\omega_0^4} e^{-4\lambda\tau}\right)^{-1}. \quad (9)$$

Moreover, the power from the oscillated charge reads

$$P_{\text{em}} = \frac{e^2}{3c^3} \omega_0^4 e^{-2\lambda\tau} \left(1 + \frac{1}{2}\tilde{\beta}_0^2 e^{-2\lambda\tau} + \frac{3}{32}\tilde{\beta}_0^4 e^{-4\lambda\tau}\right). \quad (10)$$

Now we want to compare our results based on the LD equation with the classical relativistic damping oscillation equation of a harmonic oscillator directly comes from the Newtonian case without the term $\mathcal{O}(b/c^2)$ [32]

$$\ddot{\xi} - b\xi^{(3)} = -\gamma\omega_0^2 \xi. \quad (11)$$

Our purpose here is to see the effects with and without the damping term $\mathcal{O}(b/c^2)$. This term is introduced in the LD equation but not appears in the “textbook” equation. Similarly we can still have the same zeroth-order equation (5) but the first-order equation shows λ as

$$\lambda = \frac{1}{2}b\omega^2 \left(1 + \frac{1}{4}\tilde{\beta}_0^2 e^{-2\lambda\tau} - \frac{1}{32}\tilde{\beta}_0^4 \frac{\omega^2}{\omega_0^2} e^{-4\lambda\tau}\right)^{-1}, \quad (12)$$

which is less different from the Newtonian solution for small $\tilde{\beta}_0$ that $\lambda = b\omega_0^2/2$. On the other hand, the emitted power from the oscillated damping charge is the same as that from LD equation (1) to the order of $\tilde{\beta}_0^4$.

The above results can be showed in the power spectrum profiles for a harmonically bound charge. The power radiated from the oscillator per unit frequency is

$$\frac{dW}{d\omega'} \propto \frac{1}{(\omega' - \omega^*)^2 + \lambda^2}, \quad (13)$$

where the “center frequency” $\omega^* = \omega$ satisfies equation (5), and 2λ from equation (9) or (12) is the full width at half maximum (FWHM) in the damping extended spectrum. Keep in mind in the classical Newtonian case we have $\omega^* = \omega_0$ and $\lambda = b\omega_0^2/2$, while they changes in the relativistic cases. Figure 1 gives the results for $\tilde{\beta}_0 = 0.5$ and 0.8. The FWHM is broader for the spectrum based on the LD equation compared to the electron damping equation without the term $\mathcal{O}(b/c^2)$. Also, FWHM increases with increasing of $\tilde{\beta}_0$ for the LD equation case, but decrease for the case of equation (9). That is the main difference caused by the term $\mathcal{O}(b/c^2)$. Therefore we have provided a method in principle to assess the validity and properties of LD equation. Note that here we only calculate our results to the order of $\mathcal{O}(b/c^2)$. The calculation to a higher order with shows the level shift of the center frequency $\Delta\omega^* \propto b^2$.

III. SYNCHROTRON RADIATION IN A STRONG FIELD

A. Energy Emission

The general LD equation reads

$$ma^\alpha = F_{\text{ext}}^\alpha + \frac{2e^2}{3c^3}(a^\alpha - c^{-2}u^\alpha a_\beta a^\beta), \quad (14)$$

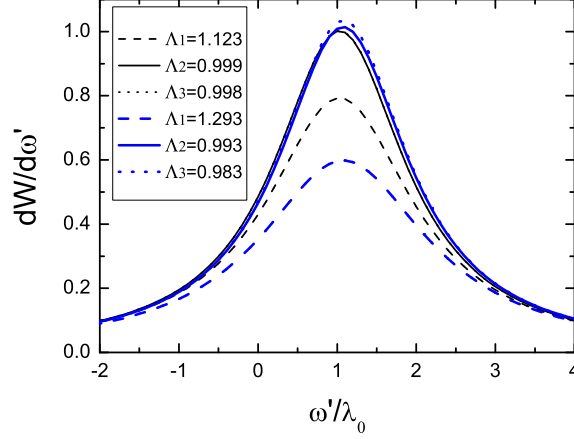


FIG. 1: Power spectrum for a harmonically bound particles damped by radiation reaction based on the original LD equation (dashed lines), damping equation (11) (solid lines) and modified LD equation (dotted lines) for $\tilde{\beta}_0 = 0.5$ (black lines) and $\tilde{\beta}_0 = 0.8$ (blue lines) with an arbitrary unit of $dW/d\omega'$. We adopt $\lambda_0 = b\omega_0^2/2$, $\Lambda_1 = \lambda_1/\lambda_0$ with λ_1 being the half FWHM based on the LD equation, $\Lambda_2 = \lambda_2/\lambda_0$ based on the damping equation (11) and $\Lambda_3 = \lambda_3/\lambda_0$ on the modified LD equation.

with the 0-component equation (i.e., the energy equation) to be

$$m\ddot{t} = F_{\text{ext}}^0 + \frac{2e^2}{3c^3}t^{(3)} - \frac{2e^2}{3c^5}\dot{t}a_\beta a^\beta. \quad (15)$$

Since the electromagnetic power emission of a charge has the covariant Larmor formula

$$P_{\text{em}} = \frac{2e^2}{3c^3}a_\beta a^\beta, \quad (16)$$

then we rewrite the energy equation (15) as

$$mc^2 \frac{d\gamma}{dt} = \frac{2e^2}{3c} \frac{d}{dt} \left(\frac{d\gamma}{d\tau} \right) - P_{\text{em}}, \quad (17)$$

which shows the change of particle total kinetic energy is not only due to the electromagnetic emission P_{em} , but also an extra radiation damping term $d\gamma/dt$. Such an energy extraction process is different to the classical result. In other words, the totally electromagnetic energy emission from a charge particle can be actually different to the total kinetic energy release $mc^2(\gamma - 1)$ if the electron goes to a non relativistic final state $\gamma = 1$. This can be considered one of the important features based on the LD equation. Let us study it more detailed.

Particles accelerated by a magnetic field $\mathbf{B}$ will radiate as the synchrotron radiation if the particles are relativistic. The motion of particles in 2-dimensional (2D) plane perpendicular to the $\mathbf{B}$ direction together with the energy equation reads [33]

$$m\ddot{\xi}_1 - \frac{2}{3} \frac{e^2}{c^3} \{ \xi_1^{(3)} - c^{-2} \dot{\xi}_1 (\ddot{\xi}_1^2 + \ddot{\xi}_2^2 - c^2 \dot{t}^2) \} = \frac{e}{c} \dot{\xi}_2 B, \quad (18)$$

$$m\ddot{\xi}_2 - \frac{2}{3} \frac{e^2}{c^3} \{ \xi_2^{(3)} - c^{-2} \dot{\xi}_2 (\ddot{\xi}_1^2 + \ddot{\xi}_2^2 - c^2 \dot{t}^2) \} = -\frac{e}{c} \dot{\xi}_1 B, \quad (19)$$

$$m\ddot{t} - \frac{2}{3} \frac{e^2}{c^3} \{ t^{(3)} - c^{-2} \dot{t} (\ddot{\xi}_1^2 + \ddot{\xi}_2^2 - c^2 \dot{t}^2) \} = 0. \quad (20)$$

Here for simplicity we assume the magnetic field is uniform, and constraint the electron motion in the plane. We will discuss the 3-dimensional (3D) motion in the last section, since the 3D motion is very similar to the 2D case. Thus

because of $c^2\dot{t}^2 - (\dot{\xi}_1^2 + \dot{\xi}_2^2) = c^2$ in 2D motion, we can assume the solution has the form:

$$\frac{d\xi_1}{d\tau} = c \sinh q \cos \theta, \quad (21)$$

$$\frac{d\xi_2}{d\tau} = c \sinh q \sin \theta, \quad (22)$$

$$c \frac{dt}{d\tau} = c \cosh q, \quad (23)$$

and equations (18)-(20) become

$$\dot{q} - b\ddot{q} + b \sinh q \cosh q \dot{\theta}^2 = 0, \quad (24)$$

$$\dot{\theta} - 2b\dot{q}\dot{\theta} \coth q - b\ddot{\theta} = -\frac{Be}{mc} = -\omega_0. \quad (25)$$

Here we define $Be/(mc) = \omega_0$, which is the frequency of gyration in the nonrelativistic case, or say the proper angular frequency $d\theta/d\tau$. If we take a new variable $\tilde{\tau} = \tau \cdot b\omega_0^2$, and hereafter still write $dq/d\tilde{\tau}$ and $d\theta/d\tilde{\tau}$ as $\dot{q}$ and $\dot{\theta}$ for simplicity, thus equation (24) and (25) are now written as

$$\dot{q} - \eta\ddot{q} + \eta \sinh q \cosh q \dot{\theta}^2 = 0, \quad (26)$$

$$\dot{\theta} - 2\eta \coth q \dot{q}\dot{\theta} - \eta\ddot{\theta} = -\eta^{-1/2}, \quad (27)$$

where $\eta = b^2\omega_0^2$ is a small dimensionless factor. The total emitted electromagnetic power is

$$P_{\text{em}} = \frac{2e^2}{3c^3} a_\eta a^\eta = \frac{2e^2}{3c} \omega_0^2 \eta (\dot{q}^2 + \dot{\theta}^2 \sinh^2 q). \quad (28)$$

From equation (26) we have

$$\dot{\theta}^2 = \frac{\eta\ddot{q} - \dot{q}}{\eta \sinh q \cosh q}, \quad (29)$$

thus equation (28) becomes

$$P_{\text{em}} = \frac{2e^2}{3c} \omega_0^2 \left(\eta \dot{q}^2 + \frac{\eta\ddot{q} - \dot{q}}{\cosh q} \sinh q \right). \quad (30)$$

The total emitted energy can be calculated as the integral of power over all the time $t = 0$ to infinity

$$\begin{aligned} E_{\text{em}} &= \int_0^\infty P_{\text{em}} dt = (b\omega_0^2)^{-1} \int_0^\infty P_{\text{em}} \cosh q d\tilde{\tau} = (b\omega_0^2)^{-1} \int_0^\infty P_{\text{em}} \cosh q \dot{q}^{-1} dq \\ &= -mc^2 \int_{q(0)}^0 \sinh q dq + mc^2 \eta \int_{q(0)}^0 (\dot{q} \cosh q + \ddot{q} \sinh q / \dot{q}) dq \end{aligned} \quad (31)$$

where we define E_0 and E_η as

$$E_0 = -mc^2 \int_{q(0)}^0 \sinh q dq = mc^2(\gamma_0 - 1) \simeq mc^2\gamma_0, \quad (32)$$

$$E_\eta = mc^2 \eta \int_{q(0)}^0 (\dot{q} \cosh q + \ddot{q} \sinh q / \dot{q}) dq = mc^2 \eta \int_{q(0)}^0 d(\dot{q} \sinh q) = -mc^2 \eta \dot{q} \sinh q_0|_{\tau=0}, \quad (33)$$

E_0 is the zeroth-order totally emitted energy from the charged particles.

Now we need to calculate the higher order energy emission E_η using the perturbation method. The solution (q, θ) can be taken as the expansion series of η and $\eta^{1/2}$:

$$q = q_0 + \eta q_1 + \eta^2 q_2 + \eta^3 q_3 + \dots, \quad (34)$$

$$\theta = -\tau \eta^{-1/2} + \theta_1 \eta^{1/2} + \theta_2 \eta^{3/2} + \theta_3 \eta^{5/2} + \dots \quad (35)$$

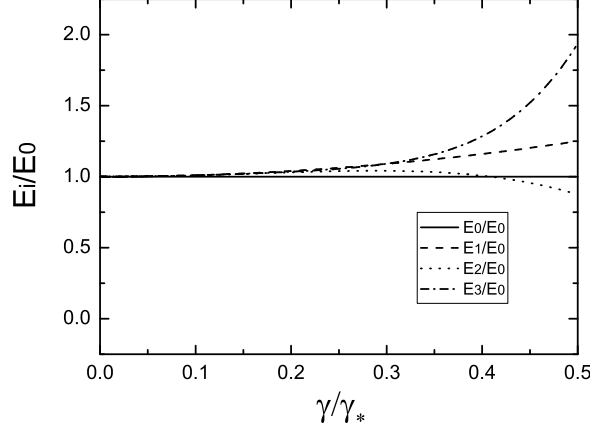


FIG. 2: Electron totally emitted energy in a strong field including the high-order calculation as in solutions (41) to (44), where we introduce the critical Lorentz factor γ_* to satisfy $\eta\gamma_*^2 = 1$. The zeroth-order energy emission $E_0 = mc^2(\gamma_0 - 1) \simeq mc^2\gamma_0$, and E_i is the energy to the i -th order, $E_i = E_0 + \sum_{k=1}^i E_\eta^k$.

Combining equations (26) and (27) and the solution form (34) and (35), the derivative of the solution coefficients q_i and θ_i at the initial point $\tau = 0$ can be calculated as (more details about the solutions using the perturbation method can be seen in Appendix A)

$$\dot{q}_0 = -\sinh q_0 \cosh q_0|_{\tau=0} = -\beta_0\gamma_0^2, \quad (36)$$

$$\dot{q}_1 = \sinh q_0 \cosh q_0 (\sinh^2 q_0 + 5 \cosh^2 q_0)|_{\tau=0} = \beta_0(\beta_0^2 + 5)\gamma_0^4, \quad (37)$$

$$\begin{aligned} \dot{q}_2 &= -(\sinh^4 q_0 + 30 \sinh^2 q_0 \cosh^2 q_0 + 37 \cosh^4 q_0) \sinh q_0 \cosh q_0|_{\tau=0} \\ &= -(\beta_0^4 + 30\beta_0^2 + 37)\beta_0\gamma_0^6, \end{aligned} \quad (38)$$

...

and

$$\dot{\theta}_1 = 2 \cosh^2 q_0|_{\tau=0}, \quad (39)$$

$$\dot{\theta}_2 = -2 \cosh^2 q_0 (3 \sinh^2 q_0 + 7 \cosh^2 q_0)|_{\tau=0}, \quad (40)$$

...

Therefore, we obtain the high order energy emitted by the charged particles as

$$E_\eta^{(1)} = mc^2\eta\beta_0^2\gamma_0^3, \quad (41)$$

$$E_\eta^{(2)} = -mc^2\eta^2\beta_0^2(\beta_0^2 + 5)\gamma_0^5, \quad (42)$$

$$E_\eta^{(3)} = mc^2\eta^3\beta_0^2(\beta_0^4 + 30\beta_0^2 + 37)\gamma_0^7, \quad (43)$$

...

$$E_\eta^{(i)} \propto (-1)^{i-1}\eta^i\gamma_0^{2i}mc^2\gamma_0 = (-1)^{i-1}(b\omega_0\gamma_0)^{2i}mc^2\gamma_0. \quad (44)$$

Figure 2 gives the electron totally emitted energy in a strong magnetic field to the third order. Note that in Figure 2 the perturbation method is only valid for $\gamma/\gamma_* \leq 0.3$ or $\eta\gamma^2 \leq 0.3$, or the series of (34) and (35) will be divergent. As a result we consider the perturbation method to be valid only if $E_\eta^{(i+1)}/E_\eta^{(i)} < 1$ or $\eta\gamma_0^2 \ll 1$. For relativistic electrons in strong magnetic fields we require $b\omega_0\gamma_0 < 1$ or

$$B_{12}\gamma_{0,4} < 1, \quad (45)$$

where $B_{12} = B/10^{12}\text{G}$ and $\gamma_{0,4} = \gamma_0/10^4$. This requirement can only be violated in the magnetic neutron star or magnetar environment where the magnetic field is extremely strong. For a typical magnetic neutron star with $B \sim 10^{12}\text{ G}$, we conclude the electron “increased” energy emission can be expected if the LD equation is correct.

Perturbation method for electron motion in a strong field has been used by some other authors [27–30, 34–36]. Shen [27] and Nelson and Wasserman [30] focused on the solutions with a lowest order correction and discussed the spectrum $P(\omega)$ and polarization $I(\omega)$ properties of synchrotron radiation. However, since the synchrotron timescale is extreme short in very strong fields, it is difficult to detect the radiation spectrum expect for the spectrum from a continuous electron sources. We will go back to discuss the radiation from a group of electron later, but with much more attention to the electrons with $B\gamma_0 > 10^{16}$ G instead. On the other hand, since the short timescale for synchrotron radiation, the electrons with a 3-dimensional initial velocity will quickly move along the magnetic field lines, and emit energy as the “curvature radiation”. Although the electrons are still damped along the curved field lines [28, 29], the power emitted in this case is much lower than the almost pulsed synchrotron radiation due to a much longer timescale. Thus the electron motion itself rather than radiation process is more important compared to the synchrotron process, and we do not discuss the curvature radiation in this paper.

Herrera [34, 35] and Lubart [36] demonstrated the trajectories of damped electron in strong fields even to high order terms, but they did not calculate the related power and energy radiation, which are the important part in our paper. In fact we can also calculate the electron motion as the function of proper time τ (or more exactly to say, $\tilde{\tau}$) besides the emission energy as the expansion series of $\eta\gamma_0^2$. The zeroth-order equation shows

$$\dot{q}_0 + \sinh q_0 \cosh q_0 = 0, \quad (46)$$

or

$$\beta_0(\tilde{\tau}) = \beta_0(0)e^{-\tilde{\tau}}, \quad (47)$$

$$\cosh q_0(\tilde{\tau}) = \frac{1}{\sqrt{1 - \beta_0^2 e^{-2\tilde{\tau}}}}, \quad (48)$$

$$\sinh q_0(\tilde{\tau}) = \frac{\beta_0 e^{-\tilde{\tau}}}{\sqrt{1 - \beta_0^2 e^{-2\tilde{\tau}}}}, \quad (49)$$

which is the same solution as the classical solution

$$\left(\frac{\gamma_0(\tau) - 1}{\gamma_0(\tau) + 1} \right) = \left(\frac{\gamma_0(0) - 1}{\gamma_0(0) + 1} \right) e^{-2b\omega_0^2 t}. \quad (50)$$

Next we have the solution of θ_1

$$\dot{\theta}_1 \eta^{1/2} = 2b\omega_0 \frac{e^{2\tilde{\tau}}}{e^{2\tilde{\tau}} - 1 + [\gamma_0(0)]^{-2}}, \quad (51)$$

for $\tilde{\tau} \rightarrow 0$, $\dot{\theta}_1 \eta^{1/2} \rightarrow 2b\omega_0$. Figure 3 gives the solution of q_i and θ_i to the second order term, or say the fourth order of b .

As a result, even some of the results in this section have been presented fragmentarily in previous literature, we try to give a unified presentation here. Furthermore, we next go to discuss the electron motion and radiation beyond the perturbation case $B\gamma < 10^{16}$ G.

B. Another Case: Extreme Strong Magnetic Fields

So far we discussed the energy emission of charged particles in the case $\eta\gamma^2 < 1$ or for electrons $B\gamma < 10^{16}$ G, which allows us to use the perturbation method. However, in the extreme case such as the relativistic electrons in the neutron star environment, the electron performance can be very different to that in last section based on the LD equation. In Section III A the power emission P_{em} to be perturbed as

$$P_{\text{em}} = \frac{2e^2}{3c} \omega_0^2 [\sinh^2 q + \eta(\dot{q}_0^2 - 2 \sinh^2 q \dot{\theta}_1) + \dots], \quad (52)$$

and we can see that P_{em} is less than the classical value $P_{\text{em,cl}} \propto \beta_0 \gamma_0^2$,

$$P_{\text{em}} = \frac{2e^2}{3c} \omega_0^2 \sinh^2 q + \frac{2e^2}{3c} \omega_0^2 \eta \cosh^2 q_0 (\sinh^2 q_0 - 4 \sinh^2 q) \quad (53)$$

$$< \frac{2e^2}{3c} \omega_0^2 \sinh^2 q = P_{\text{em,cl}}. \quad (54)$$

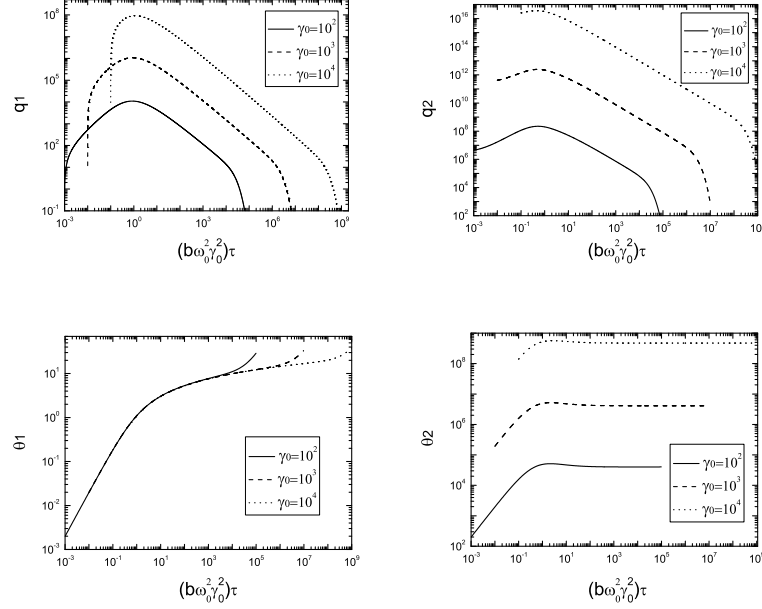


FIG. 3: The coefficients q_i and θ_i of series (34) and (35) to the second order.

In this section we want to show that for extreme strong fields the power emission has $P_{\text{em}} \ll P_{\text{em,cl}}$, while the exact value of P_{em} depends on the initial acceleration condition as well as the velocity condition.

If $b\omega_0\gamma_0 > 1$, we first consider the analytic limit for $b\omega_0\gamma_0 \gg 1$. Taking $\epsilon = (b\omega_0\gamma_0)^{-1}$, we have the motion equations as

$$\ddot{q} - \sinh q \cosh q \dot{\theta}^2 - \epsilon \dot{q} = 0, \quad (55)$$

$$\ddot{\theta} + 2 \coth q \dot{q} \dot{\theta} - \epsilon \dot{\theta} - \frac{1}{\gamma_0} \epsilon^{3/2} = 0. \quad (56)$$

Different to Section III A, here we rewrite $\dot{q} := \gamma_0^{-2} dq/d\tilde{\tau}$, $\dot{\theta} := \gamma_0^{-2} d\theta/d\tilde{\tau}$, $\ddot{q} := \gamma_0^{-4} d^2q/d\tilde{\tau}^2$, $\ddot{\theta} := \gamma_0^{-4} d^2\theta/d\tilde{\tau}^2$, and $\tau^* := \gamma_0^2 \tilde{\tau} = b\omega_0^2 \gamma_0^2 \tau$ in this section. Then q and θ can be expanded as the series of ϵ and $\epsilon^{1/2}$

$$q = q_0 + \epsilon q_1 + \epsilon^2 q_2 + \epsilon^3 q_3 + \dots \quad (57)$$

$$\theta = \frac{1}{\gamma_0} (\epsilon^{1/2} \theta_1 + \epsilon^{3/2} \theta_2 + \epsilon^{5/2} \theta_3 + \dots). \quad (58)$$

We expand equations (55) and (56) to the second order q_2 and θ_2 , i.e.,

$$\ddot{q}_0 = 0, \quad (59)$$

$$\ddot{q}_1 - \dot{q}_0 + \frac{A_0}{\gamma_0} \dot{\theta}_1^2 = 0, \quad (60)$$

$$\ddot{q}_2 - \dot{q}_1 + \frac{A_1}{\gamma_0} \dot{\theta}_1^2 + 2 \frac{A_0}{\gamma_0} \dot{\theta}_1 \dot{\theta}_2 = 0. \quad (61)$$

...

and

$$\ddot{\theta}_1 + 2\dot{q}_0 B_0 \dot{\theta}_1 = 0, \quad (62)$$

$$\ddot{\theta}_2 - \dot{\theta}_1 - 1 + 2\dot{q}_1 \dot{\theta}_1 B_0 + 2\dot{q}_0 \dot{\theta}_1 B_1 + 2\dot{q}_0 \dot{\theta}_2 B_0 = 0. \quad (63)$$

...

Now we should face the problem which does not exist in the perturbation case with $B\gamma_0 < 10^{16}$ G: what is the initial acceleration condition $\dot{q}$ and $\dot{\theta}$, which is necessary to know in the second order equations? If we take no initial

acceleration $\dot{q}(0) = 0$, we have the perturbation solutions:

$$\dot{q}_0 \equiv 0, \quad \dot{\theta}_1 \equiv 0, \quad \dot{q}_1 \equiv 0, \quad \dot{q}_2 \equiv 0, \quad \dot{\theta}_2 = \tau^*, \quad \dot{q}_2 \equiv 0, \quad \dot{q}_3 = \frac{A_0}{3\gamma_0^2}(\tau^*)^3. \quad (64)$$

Thus the emission electron power as the function of proper time is

$$P_{\text{em}} = \frac{2e^2}{3c} \omega_0^2 \sinh^2 q_0 \left(\frac{\tau}{b} \right)^2, \quad (65)$$

where the rescaled proper time τ^* is estimated

$$\tau^* \sim \frac{q}{\dot{q}} \sim \frac{q_0}{\epsilon^3 \dot{q}_3} \sim \frac{q_0}{\epsilon^3 (\tau^*)^3}. \quad (66)$$

or

$$(\tau^*)^4 \sim b^6 \omega_0^6 \gamma_0^6 q_0, \quad (67)$$

and we have

$$\left(\frac{\tau}{b} \right)^4 \sim \frac{1}{(b\omega_0\gamma_0)^2} q_0 \ll 1. \quad (68)$$

Therefore according to equation (65) the emission power is much less than the classical value as

$$P_{\text{em}} \ll \frac{2e^2}{3c} \omega_0^2 \sinh^2 q_0 = P_{\text{em,cl}}. \quad (69)$$

The first order of total energy emission is

$$E_\eta \propto \dot{q} \propto \epsilon^3 (\tau^*)^3|_0 = 0, \quad (70)$$

or the totally emitted energy is $E \simeq E_0$ to the third order.

On the other hand, if the electrons are injected or created in the magnetic fields with initial acceleration $dq/d\tau^*|_0 = Q$, then the electron power as

$$P_{\text{em}} = mc^2 b^3 \omega_0^4 \gamma_0^4 Q^2. \quad (71)$$

We estimate $b\omega_0^2 \gamma_0^2 Q = dq/d\tau \sim \gamma^{-1}(d\gamma/d\tau)$, then we have

$$\sqrt{\frac{P_{\text{em}}}{P_{\text{em,cl}}}} \sim \sqrt{\frac{b^3 \omega_0^4 \gamma_0^4 Q^2}{b\omega_0^2 \gamma_0^2 \beta^2}} \simeq b\omega_0 \gamma_0 |Q| \sim \left| \frac{d\gamma}{d\tau} \right| (\omega_0 \gamma_0^2)^{-1} \ll \frac{b}{\gamma_0} \left| \frac{d\gamma}{d\tau} \right| \sim \left(\frac{b}{\tau_{\text{acc}}} \right) \ll 1. \quad (72)$$

We still have $P_{\text{em}} \ll P_{\text{em,cl}}$ and the first order to total energy emission is

$$E_\eta = -\eta Q \beta_0 \gamma_0^3 \sim mc^2 b \left| \frac{d\gamma}{d\tau} \right| \sim E_0 \left(\frac{b}{\tau_{\text{acc}}} \right) \ll mc^2 \gamma_0. \quad (73)$$

Let us make a brief summary for Section III, i.e., the electron radiation properties based on the LD equation. One of the difference between the results in Section III and the classical synchrotron theory is that the total electron emission can be different to the kinematic change $mc^2(\gamma - 1)$, which means the damping term also contributes to the total energy emission. However, the energy emission is still very close to $E_0 = mc^2(\gamma - 1)$, as we calculated in two cases $E_\eta = E_{\text{em}} - E_0 \sim \mathcal{O}(\eta\gamma^2)$ for $B\gamma < 10^{16}$ G, and also $E_\eta \ll E_0$ for $B\gamma > 10^{16}$ G.

The radiation power P_{em} , on the other hand, can be significantly lower than the classical value $P_{\text{em,cl}} \approx \beta^2 \gamma^2$. We have $P_{\text{em}} - P_{\text{em,cl}} \sim \mathcal{O}(\eta)$ for the perturbation case, but $P_{\text{em}} \ll P_{\text{em,cl}}$ for the extreme strong fields. However, the value of power P_{em} should also depends on the initial acceleration condition of electrons in extreme strong fields.

Since the electron emission process in strong fields is too fast to be directly detected even in principle, the better way is to apply the theory to a continuous electron source in a strong field and see the radiation power and energy by such a group of electron. The simplest way is to consider a group of electron with the similar Lorentz factor, but the more commonly considered distribution is the power law distribution $n_e(\gamma) \propto \gamma^{-p}$. We do not want to calculate the results here, which cannot be obtained analytically based on the LD equation. Keeping in mind the basic results by a single electron mentioned above should not be changed for the continuous electron source radiation, we will calculate the group electron radiation based on the modified LD equation in Section IV B, and go back to discuss them together.

IV. MODIFIED LORENTZ-DIRAC DAMPING EQUATION

As mentioned in the introduction section, the original LD equation meets difficulties because it unusually involves the derivative of the acceleration vector. A third-order equation not only requires the particles position and velocity at the initial proper time, but also the initial acceleration condition if the perturbation method is invalid. The two main problems, so called “runaway” and “precelebration” solutions are like that, for an arbitrary choice of non-zero initial acceleration, the acceleration grows exponentially with time; while if we want to eliminate this problem, the information before the initial time $\tau = 0$ should be included into the solution, which has a contradiction with elementary ideas of causality. There have been several similar possible interpretations to eliminate the problems. We follow the comments and ideas from Poisson [4] and Rohrlich [11], the original LD can be modified to a new relativistic equation

$$ma^\alpha = F_{\text{ext}}^\alpha + b(\delta^\alpha_\beta + c^{-2}u^\alpha u_\beta)\dot{F}_{\text{ext}}^\beta, \quad (74)$$

which no longer involves $\dot{a}$ but the derivative of force $\dot{F}$ instead. The above equation (74) does not give rise to runaway and precelebration solutions. Our purpose in this section is to calculate the energy emission from accelerated charges based on the modified equation (74) to see the effects of radiation damping in this case, and shows the differences between the results based on equation (74) and the original one (14) for potential detection in the future to assess which one is correct.

A. Harmonically Bound Charges

For the harmonic oscillators the force and its derivative can be written as

$$F = -\gamma k\xi \simeq -\left(1 + \frac{1}{2}\tilde{\beta}^2 - \frac{1}{8}\tilde{\beta}^4\right)k\xi, \quad (75)$$

$$\dot{F} = -\left(\tilde{\beta}\dot{\tilde{\beta}} - \frac{1}{2}\tilde{\beta}^3\dot{\tilde{\beta}}\right)k\xi - \left(1 + \frac{1}{2}\tilde{\beta}^2 - \frac{1}{8}\tilde{\beta}^4\right)k\dot{\xi}, \quad (76)$$

where we assume the non-relativistic case $\tilde{\beta} < 1$ or $\beta < 1/\sqrt{2}$. Thus the 1-dimensional space component of modified equation (74) reads

$$\ddot{\xi} = -\left(1 + \frac{1}{2}\tilde{\beta}^2 - \frac{1}{8}\tilde{\beta}^4\right)\omega_0^2\xi + b\omega_0^2(1 + \tilde{\beta}^2)\frac{\dot{F}}{m}. \quad (77)$$

We still take the solution as the form of (3) in Section II. The zeroth-order result (5) is unchanged. However, the first order equation gives the value of λ as

$$\lambda = \frac{1}{2}b\omega_0^2\left(1 + \frac{1}{4}\tilde{\beta}_0^2\frac{\omega^2}{\omega_0^2}e^{-2\lambda\tau} - \frac{7}{64}\tilde{\beta}_0^4\frac{\omega^4}{\omega_0^4}e^{-4\lambda\tau}\right)\left(1 + \frac{1}{4}\tilde{\beta}_0^2e^{-2\lambda\tau} - \frac{1}{32}\tilde{\beta}_0^4\frac{\omega^2}{\omega_0^2}e^{-4\lambda\tau}\right)^{-1} \quad (78)$$

Taking the approximation expression (6) for the zeroth-order, we have

$$\lambda \simeq \frac{1}{2}b\omega_0^2, \quad (79)$$

which is unchanged compared to the classical damping formula (12) without the term $\mathcal{O}(b/c^2)$.

We also plot the power spectrum of an oscillator based on the modified LD equation in Figure 1 and compare it with those in Section II. The power spectrum is much closer to the classical spectrum based on equation (11) without $\mathcal{O}(b/c^2)$, and the FWHM decreases with increasing the oscillator's velocity $\tilde{\beta}_0$. This property is important to distinguish the modified LD equation to the original one in the harmonically oscillator case, and can be expected to be detected by experiment in principle to see which electron motion is real to fit the real spectrum profile.

B. Synchrotron Radiation

The energy equation (i.e., 0-component of equation (74)) reads

$$m\ddot{t} = F_{\text{ext}}^0 + b\dot{F}_{\text{ext}}^0 + bc^{-2}tu_\alpha\dot{F}_{\text{ext}}^\alpha \quad (80)$$

and we have

$$ma_\mu a^\mu = F_{\text{ext}}^\alpha a_\alpha + b\dot{F}_{\text{ext}}^\alpha a_\alpha. \quad (81)$$

For the Lorentz force in a pure magnetic field, the force and its derivative satisfy

$$u_\alpha F_{\text{ext}}^\alpha = 0, \quad (82)$$

and also

$$a_\alpha \dot{F}_{\text{ext}}^\alpha = 0, \quad (83)$$

therefore the energy equation can be written as

$$mc^2 \ddot{t} = -bmta_\mu a^\mu, \quad (84)$$

for $F^0 = 0$ (i.e., no electric acceleration), or the familiar form

$$mc^2 \frac{d\gamma}{dt} = -P_{\text{em}}. \quad (85)$$

This energy equation is same as the classical one in the radiative process without damping. In other words, the total energy emission is $mc^2(\gamma_0 - 1)$, which is consistent with the classical conclusion but different with the results from the original LD equation. However, the expression for the power P_{em} here will be different with radiation damping. Next we solve the equation (74) in the particular case, which can be written as

$$\ddot{\xi}_1 = \omega_0 \dot{\xi}_2 + b\omega_0 \ddot{\xi}_2 + b\omega_0 c^{-2} \dot{\xi}_1 (\dot{\xi}_1 \ddot{\xi}_2 - \dot{\xi}_2 \ddot{\xi}_1), \quad (86)$$

$$\ddot{\xi}_2 = -\omega_0 \dot{\xi}_1 - b\omega_0 \ddot{\xi}_1 + b\omega_0 c^{-2} \dot{\xi}_2 (\dot{\xi}_1 \ddot{\xi}_2 - \dot{\xi}_2 \ddot{\xi}_1), \quad (87)$$

$$\ddot{t} = b\omega_0 c^{-2} \dot{t} (\dot{\xi}_1 \ddot{\xi}_2 - \dot{\xi}_2 \ddot{\xi}_1), \quad (88)$$

if we still constrain the electron motion in the plane perpendicular to the magnetic field lines. We use the variables as in (21) to (23). Similarly we adopt $\eta = b^2 \omega_0^2$, $\dot{q} = (b\omega_0^2) dq/d\tau$, $\dot{\theta} = (b\omega_0^2) d\theta/d\tau$ and $\tilde{\tau} = b\omega_0^2 \tau$. Then the equations (86) to (88) becomes

$$\dot{\theta} + \sqrt{\eta} \dot{q} \coth q = -\frac{1}{\sqrt{\eta}}, \quad (89)$$

$$\dot{q} = \sqrt{\eta} \dot{\theta} \sinh q \cosh q. \quad (90)$$

The solution of equations (89) and (90) can be obtained analytically as

$$\frac{(\sinh q)^{1+\eta}}{\cosh q} = \frac{(\sinh q_0)^{1+\eta}}{\cosh q_0} e^{-\tilde{\tau}}. \quad (91)$$

Compared to the classical result or the zero-order solution (47) of the original LD equation in Section III, we have

$$\begin{aligned} \frac{\Delta\gamma}{\gamma} &= \frac{\eta}{2} \left[1 - \left(\frac{\gamma}{\gamma_0} \right)^2 e^{-\tilde{\tau}} + \ln \left(\frac{\gamma}{\gamma_0} \right) (2\gamma^2 - 1) \right] \\ &\rightarrow \frac{\eta}{2} \left[1 + \ln \left(\frac{\gamma_0}{\gamma} \right) \right] \propto \eta, \end{aligned} \quad (92)$$

where we take $\Delta\gamma$ as the difference between those derived from (47) and (91) as the function of $\tilde{\tau}$, and always take the general case $\eta \ll 1$.

¹ The condition $\eta \ll 1$ can always be satisfied except in extremely strong magnetic field $B > 10^{16}$ G, which can only be reached inside the newborn magnetar [37].

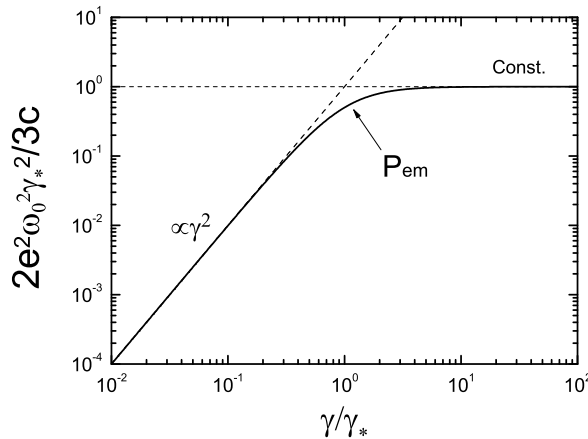


FIG. 4: Synchrotron power emission of charged particle based on the modified LD equation solution (93), where γ_* is chosen at $\eta\gamma_*^2 = 1$ as in Fig. 2, and $\omega_0 = Be/mc$.

The emission power can be derived based on the solution (91) as

$$P_{\text{em}} = \frac{2e^2}{3c} \omega_0^2 \frac{\sinh^2 q}{1 + \eta \cosh^2 q} = \frac{2e^2}{3c} \omega_0^2 \frac{\beta^2 \gamma^2}{1 + \eta \gamma^2}, \quad (93)$$

Therefore the power P_{em} is much lower than the classical radiation if $\eta\gamma^2 \gg 1$. The limit power emission for strong fields keeps as a constant

$$P_{\text{em}} = \frac{2e^2}{3c} \frac{\omega_0^2}{\eta} \beta^2 = mc^2 \beta^2 b^{-1} \ll mc^2 b \omega_0^2 \beta^2 \gamma^2 = P_{\text{em,cl}} \quad (94)$$

Note that this constant solution is very similar to that of expression (72), whose only difference is that (72) needs the further information about initial acceleration. Figure 4 shows the difference of synchrotron power radiation between equation (93) and the classical result.

Next we consider a continuous electron source with the power law distribution $n_e(\gamma) = n_0 \gamma^{-p}$. If we assume the strong field environment is optically thin to its synchrotron radiation, the total electron power is

$$\begin{aligned} P_{\text{tot}} &= \int P_{\text{em}} n_e(\gamma) d\gamma = \frac{2e^2}{3c} \omega_0^2 n_0 \int_{\gamma_{\min}}^{\gamma_{\max}} \frac{\beta^2 \gamma^{2-p}}{1 + \eta \gamma^2} d\gamma \\ &= \frac{2e^2}{3c} \omega_0^2 n_0 \gamma_*^{3-p} \int_{\gamma_{\min}/\gamma_*}^{\gamma_{\max}/\gamma_*} \frac{x^{2-p}}{1 + x^2} dx, \end{aligned} \quad (95)$$

where $x = \gamma/\gamma_*$ with γ_* satisfies $\eta\gamma_*^2 = 1$. Usually we take $\gamma_{\min}/\gamma_* \simeq 0$, and if we assume $\gamma_{\max} \gg \gamma_*$, then the total electron power radiation for the modified LD equation $P_{\text{tot}} \propto \text{Const.}$ as a function of p will be much less than the classical result where $P_{\text{tot,cl}} \propto \gamma_{\max}^{3-p}$ for $p < 3$. This conclusion that the radiation damping decreases the total synchrotron radiation power significant in the strong fields is basically consistent with the result from the original LD equation qualitatively. However, it is easier to derive the expression of total emission P_{tot} with the modified LD equation, while P_{tot} from the original LD equation is also a function of the initial acceleration conditions.

Besides the analytic calculation, we can also use the iteration method to calculate the modified LD equation in another example. Appendix B gives this example to show the difference between LD and modified LD equations using the iteration method to $\mathcal{O}(b^2)$.

V. ELIEZER'S EQUATION

We discuss another type of “generalized” LD equation for completeness in this section. As mentioned before, the modified LD equation uses the derivative of force $\dot{F}$ instead of the derivative of acceleration in order to eliminate the

“runaway” or “precelebration” problems. Another way to deal with the problems using the opposite consideration is to introduce even higher derivatives into the original LD equation, which was first studied by Eliezer [5, 7]. Following Dirac’s method, LD equation can be actually extended to a more general form (hereafter Eliezer equation, see the Appendix B for more details):

$$\dot{v}^\alpha = b(\ddot{v}^\alpha - c^{-2}\dot{v}^2 v^\alpha) + \dot{\mathcal{D}}^\alpha + F_{\text{ext}}^\alpha, \quad (96)$$

where

$$\begin{aligned} \dot{\mathcal{D}}^\alpha = & c^{-2}v^\alpha \left\{ -\frac{3}{2}P_2\dot{v}^2 - 5P_4^v(4)_\mu\dot{v}^\mu - \frac{5}{2}P_4\ddot{v}^2 \right\} + \dots \\ & + P_2\ddot{v}^\alpha + P_4v^{4,\alpha} + \dots \end{aligned} \quad (97)$$

with $P_2, P_4 \dots$ as the parameters which need to be further determined. The Eliezer’s equation which directly extends LD equation mathematically meets several difficulties. First of all, the parameters (P_2, P_4 etc.) cannot be derived from the classical electrodynamical theory. Second, as the performance of these general equations are not well studied, we cannot know if the general equation with certain parameter space can eliminate the problems which LD equation meets. Based on the justification of Occam’s razor, we can hardly believe this equation can be true except for the special case $P_{2i} \equiv 0$ which goes back to the original LD equation. To discuss the meaning and properties of the general Eliezer’s equation is beyond the purpose of this paper. What we want to do in this section is just provide a possible way to test the Eliezer equation, or in other word, to determine the parameters of Eliezer’s equation in some particular forms.

The simplest assumption $P_{2i} \equiv 0$ switches back to the LD equation. The first extension of LD equation is that only P_2 is left in the factor $\mathcal{D}$ of equation (97), then we can rewritten the Eliezer’s equation as

$$ma^\alpha = bm(\dot{a}^\alpha - c^{-2}u^\alpha a_\beta a^\beta) + mkb^2 \frac{d}{d\tau}(2\dot{a}^\alpha/3 - c^{-2}u^\alpha a_\beta a^\beta) + F_{\text{ext}}^\alpha, \quad (98)$$

where k is the added parameter into the LD equation. Thus the energy equation is

$$mc^2\ddot{t} = -\gamma P_{\text{em}} - bk \frac{d}{d\tau}(\gamma P_{\text{em}}) + mc^2bt^{(3)} + \frac{2k}{3}mc^2b^2t^{(4)}, \quad (99)$$

or

$$\begin{aligned} E_{\text{em}} = & mc^2(\gamma_0 - 1) - mc^2\eta\dot{q}\sinh q_0|_{\tau=0} \\ & + kmc^2\eta^2\gamma_0(\dot{\theta}^2\sinh q_0 + \dot{q}^2)|_{\tau=0} - \frac{2}{3}mc^2\eta^2k(\ddot{q}\sinh q_0 + \dot{q}^2\cosh q_0)|_{\tau=0}. \end{aligned} \quad (100)$$

The electron energy release is more complicated than the original equation if $k \neq 0$.

Using the variable transformation (21) to (23) and the simplified notification above (26) and (27), we rewrite the equation of motion (98) as

$$\begin{aligned} & \dot{q} - \eta\ddot{q} + \eta\sinh q \cosh q \dot{\theta}^2 \\ & - k\eta^2 \left[\frac{2}{3}\ddot{q} - \frac{1}{3}\dot{q}^3 - (2\cosh^2 q + \sinh^2 q)\dot{q}\dot{\theta}^2 - 2\sinh q \cosh q \dot{q}\ddot{\theta} \right] = 0, \end{aligned} \quad (101)$$

$$\begin{aligned} & \dot{\theta} - \eta(2\dot{q}\dot{\theta}\coth q + \ddot{\theta}) \\ & - k\eta^2 \left[-\dot{\theta}^3\sinh^2 q + 2\dot{q}\dot{\theta}\coth q + 2\dot{q}\ddot{\theta}\coth q + \dot{q}^2\dot{\theta} + \frac{2}{3}\theta^{(3)} - \frac{2}{3}\dot{\theta}^3 \right] = -\frac{1}{\sqrt{\eta}}. \end{aligned} \quad (102)$$

The perturbation method introduces the equations of q_i and θ_i to the first order as

$$\dot{\theta}_0 = -1, \quad (103)$$

$$\dot{q}_0 = -\sinh q_0 \cosh q_0, \quad (104)$$

$$\dot{\theta}_1 = 2\cosh^2 q_0 + k(\sinh^2 q_0 + 2/3), \quad (105)$$

$$\dot{q}_1 = (\sinh^2 q_0 + 5\cosh^2 q_0)\sinh q_0 \cosh q_0 + k\sinh q_0 \cosh q_0(3\sinh^2 q_0 + 2\cosh^2 q_0 + 4/3), \quad (106)$$

and we can derive the first and second order of energy emission with the parameter k as

$$E_\eta^{(1)} = (1+k)mc^2\eta\gamma_0^3, \quad (107)$$

$$E_\eta^{(2)} = -mc^2\eta^2\gamma_0^5(6+10k+2k^2), \quad (108)$$

which directly establish the relation between the value of k and the observable quantities.

VI. CONCLUSIONS AND DISCUSSION

In this paper we study two examples of electron radiation based on the Lorentz-Dirac (LD) equation as well as its modified equations: radiation spectrum profiles of a harmonically bound charge and synchrotron radiation from a ultra relativistic electron in a strong magnetic field.

For a harmonically bound electron we compare the solutions of electron motion and the corresponding spectrum profiles based on different motion of equations.

(1) LD equation: the FWHM of its radiation spectral profile is broader compared to the electrom damping equation without the term $\mathcal{O}(b/c^2)$ (i.e., equation (11)). The FWHM increases with increasing of $\tilde{\beta}_0$, but decreases based on equation (11).

(2) Modified LD equation: the properties of radiation spectral profile from a harmonically bound electron are more similar to those based on the damping equation (11). The different profiles provide a clue in principle to assess the validity of the original LD equation and its modified form.

For the synchrotron radiation from a electron which is constrained in the plane perpendicular to the magnetic field lines, the results of electron radiation process based on different motion of equations as follows,

(1) LD equation: The power emission significantly decreases for $B\gamma_0 > 10^{16}$ G compared to the classical result $P_{\text{em,cl}} \propto \beta_0^2 \gamma_0^2$. The total emitted energy will be higher than the kinetic energy change $(\gamma_0 - 1)m_e c^2$ due to the radiation damping effects. However, the actual radiation power and energy depend on the the electron initial acceleration condition as well as the initial velocity and Lorentz factor.

(2) Modified LD equation: The totally emitted energy should be the same as the electron kinetic energy release $(\gamma_0 - 1)m_e c^2$. We derive the analytic solutions of synchrotron radiation power and electron motion, which is only depend on the initial velocity condition. The power emission significantly decreases for $B\gamma_0 > 10^{16}$ G compared to the classical result.

(3) Eliezer's equation: As the generalized LD equation, Eliezer's equation gives the electron radiation as the correction of that derived from the original LD equation with parameters. Therefore it leaves room for the correction of radiation power and total energy from those derived from the LD equation.

Compared to the case of harmonically bound charge, we emphases on the similarities of electron radiation in strong fields in the three cases. First of all we expect the radiation reaction effects in strong fields can be observed for a group of electrons in the magnetic neutron star or magnetar environments in principle. After that we point out the way to distinguish the different radiation behaviors based on the three equations and assess the validity of one of them.

So far we constrained the electron motion in the 2D surface perpendicular to the magnetic fields. Actually the results in this paper can be extended to the 3D space. If we assume the electron coordinate is (ξ_1, ξ_2, ξ_3, t) with the added ξ_3 along the magnetic field, we always have the ratio between $\dot{\xi}_3$ and $\dot{t}$ is a constant for all the electron motion of equations, i.e., $\dot{\xi}_3 = v_3 \dot{t}$ with v_3 being the constant. If we take $\sin \alpha = (1 - v_3^2/c^2)^{1/2}$, then the transformation formulae (21) to (23) can be modified as

$$\dot{\xi}_1 = c \sinh q \cos \theta, \quad (109)$$

$$\dot{\xi}_2 = c \sinh q \sin \theta, \quad (110)$$

$$\dot{\xi}_3 = v_3 (\sin \alpha)^{-1} \cosh q, \quad (111)$$

$$c\dot{t} = c (\sin \alpha)^{-1} \cosh q, \quad (112)$$

and the equations of q and θ as well as the radiation as the function of q and θ is still not changes. However, keep in mind in the 3D case we have $\cosh q = \gamma \sin \alpha$, $\sinh q = (v^2 - v_3^2)^{1/2} \gamma / c$ or in the relativistic case $\sinh q \simeq \gamma \sin \alpha$, and $dt = (\sin \alpha)^{-1} \cosh q d\tau$. Therefore, we should replace the γ_0 and $\beta_0 \gamma_0$ in equations (36) to (40) as $\gamma_0 \sin \alpha$, E_η^i in equation (41) to (44) should time $(\sin \alpha)^{2i}$. Also, the electron kinematic energy release or the zeroth-order electron energy emission is now $mc^2[\gamma - (\sin \alpha)^{-1}]$. The exact solution of P_{em} (93) based on the modified LD equation becomes

$$P_{\text{em}} = \frac{2e^2}{3c^3} \omega_0^2 \frac{(v^2 - v_3^2) \gamma^2}{1 + \eta \gamma^2 \sin^2 \alpha}, \quad (113)$$

or

$$P_{\text{em}} = \frac{2e^2}{3c} \omega_0^2 \frac{\gamma^2 \sin^2 \alpha}{1 + \eta \gamma^2 \sin^2 \alpha} \quad (114)$$

for the relativistic case.

Acknowledgements

I thank

APPENDIX A: PERTURBATION METHOD

Since η in equations (26) and (27) is a small quantity, we can solve them using the perturbation method. The solution (q, θ) can be taken as the expansion series of η and $\eta^{1/2}$. Then equations (26) and (27) become a series of equations among q_i and θ_i .

$$q = q_0 + \eta q_1 + \eta^2 q_2 + \eta^3 q_3 + \dots, \quad (\text{A1})$$

$$\theta = -\tau \eta^{-1/2} + \theta_1 \eta^{1/2} + \theta_2 \eta^{3/2} + \theta_3 \eta^{5/2} + \dots \quad (\text{A2})$$

The Taylor expansion of functions $\cosh q \sinh q$ and $\coth q$ can be write as

$$\begin{aligned} \cosh q \sinh q &= \sinh q_0 \cosh q_0 + \eta q_1 (\sinh^2 q_0 + \cosh^2 q_0) \\ &\quad + \eta^2 (q_2 \sinh^2 q_0 + q_2 \cosh^2 q_0 + 2q_1^2 \sinh q_0 \cosh q_0) \\ &\quad + \eta^3 (q_3 \sinh^2 q_0 + q_3 \cosh^2 q_0 + 4q_1 q_2 \sinh q_0 \cosh q_0 + \frac{2}{3} q_1^3 \sinh^2 q_0 + \frac{2}{3} q_1^3 \cosh^2 q_0) \\ &\quad + \eta^4 (q_4 \sinh^2 q_0 + q_4 \cosh^2 q_0 + 2q_2^2 \sinh q_0 \cosh q_0 + 4q_1 q_3 \sinh q_0 \cosh q_0 \\ &\quad + 2q_1^2 q_2 \sinh^2 q_0 + 2q_1^2 q_2 \cosh^2 q_0 + \frac{2}{3} q_1^4 \sinh q_0 \cosh q_0) + \dots \\ &:= A_0 + \eta A_1 + \eta^2 A_2 + \eta^3 A_3 + \eta^4 A_4 + \dots, \end{aligned} \quad (\text{A3})$$

$$\begin{aligned} \coth q &= \coth q_0 - \eta q_1 (\sinh q_0)^{-2} + \eta^2 [-q_2 (\sinh q_0)^{-2} + q_1^2 (\sinh q_0)^{-3} \cosh q_0] \\ &\quad + \eta^3 [-q_3 (\sinh q_0)^{-2} + 2q_1 q_2 (\sinh q_0)^{-3} \cosh q_0 - q_1^3 (\sinh q_0)^{-4} \cosh^2 q_0 + \frac{1}{3} q_1^3 (\sinh q_0)^{-2}] \\ &\quad + \eta^4 [-q_4 (\sinh q_0)^{-2} + q_2^2 (\sinh q_0)^{-3} \cosh q_0 + 2q_1 q_3 (\sinh q_0)^{-3} \cosh q_0 + \\ &\quad - 3q_1^2 q_2 (\sinh q_0)^{-4} \cosh^2 q_0 + q_1^2 q_2 (\sinh q_0)^{-2} + q_1^4 (\sinh q_0)^{-5} \cosh^3 q_0 \\ &\quad - \frac{2}{3} q_1^4 (\sinh q_0)^{-3} \cosh q_0] + \dots \\ &:= B_0 + \eta B_1 + \eta^2 B_2 + \eta^3 B_3 + \eta^4 B_4 + \dots, \end{aligned} \quad (\text{A4})$$

where A_i and B_i are introduced to notify the coefficients in the series.

Combining the expanded series (A1) to (A4) into equations (26) and (27), we have a series of equations as

$$\dot{q}_0 + A_0 = 0, \quad (\text{A5})$$

$$\dot{q}_1 - \ddot{q}_0 + A_1 - 2A_0 \dot{\theta}_1 = 0, \quad (\text{A6})$$

$$\dot{q}_2 - \ddot{q}_1 + \dot{\theta}_1^2 A_0 - 2A_1 \dot{\theta}_1 + A_2 - 2A_0 \dot{\theta}_2 = 0, \quad (\text{A7})$$

$$\dot{q}_3 - \ddot{q}_2 - 2A_0 \dot{\theta}_3 + 2A_0 \dot{\theta}_1 \dot{\theta}_2 + A_1 \dot{\theta}_1^2 - 2A_1 \dot{\theta}_2 - 2A_2 \dot{\theta}_1 + A_3 = 0, \quad (\text{A8})$$

...

and

$$\dot{\theta}_1 + 2\dot{q}_0 B_0 = 0, \quad (\text{A9})$$

$$\dot{\theta}_2 - \ddot{\theta}_1 + 2\dot{q}_1 B_0 + 2\dot{q}_0 B_1 - 2\dot{q}_0 \dot{\theta}_1 B_0 = 0, \quad (\text{A10})$$

$$\dot{\theta}_3 - \ddot{\theta}_2 + 2\dot{q}_2 B_0 + 2\dot{q}_1 B_1 + 2\dot{q}_0 B_2 - 2\dot{q}_1 \dot{\theta}_1 B_0 - 2\dot{q}_0 \dot{\theta}_1 B_1 - 2\dot{q}_0 \dot{\theta}_2 B_0 = 0, \quad (\text{A11})$$

...

If $\tau = 0$, taking the note $\dot{\theta}_0 = -1$, we can simplify the equations as

$$\dot{q}_{k+1} - \ddot{q}_k + \sum_{0 \leq i \leq j}^{i+j=k+1} (2 - \delta_{ij}) \dot{\theta}_i \dot{\theta}_j A_0 = 0, \quad (\text{A12})$$

$$\dot{\theta}_{k+1} - \ddot{\theta}_k - 2 \sum_{\substack{i+j=k \\ 0 \leq i \leq k}} \dot{q}_i \dot{\theta}_j B_0 = 0, \quad (\text{A13})$$

which has been used to calculate the energy emission E_η in Section III A.

APPENDIX B: MODIFIED LORENTZ-DIRAC EQUATION IN PERTURBATION FORM

We rewrite the modified LD equation (74) as

$$\frac{1}{m} F^\alpha = a^\alpha - b(\delta^\alpha_\beta + c^{-2} u^\alpha u_\beta) \frac{\dot{F}^\beta}{m}. \quad (\text{B1})$$

If the second term in the right hand of the above equation is much less than the first term, we can solve this equation using an iteration method. Extend equation (B1) to the second order

$$a^\alpha - b(\dot{a}^\alpha - c^{-2} u^\alpha a_\beta a^\beta) + b^2(\ddot{a}^\alpha - 3c^{-2} u^\alpha a_\delta \dot{a}^\delta - c^{-2} a^\delta a_\gamma \dot{a}^\gamma) = F_{\text{ext}}^\alpha / m. \quad (\text{B2})$$

The 2-dimensional equation in the electromagnetic $(\mathbf{E}, \mathbf{B})$ environment reads

$$\begin{aligned} & \ddot{\xi}_1 - b[\xi_1^{(3)} - c^{-2} \dot{\xi}_1(\ddot{\xi}_1^2 + \ddot{\xi}_2^2 - c^2 \dot{t}^2)] \\ & + b^2[\xi_1^{(4)} - 3c^{-2} \dot{\xi}_1(\ddot{\xi}_1 \xi_1^{(3)} + \ddot{\xi}_1 \xi_1^{(3)} - c^2 \dot{t} t^{(3)}) - c^{-2} \ddot{\xi}_1(\ddot{\xi}_1^2 + \ddot{\xi}_2^2 - c^2 \dot{t}^2)] \\ & = \frac{e}{mc} \dot{\xi}_2 B + \frac{e E_1}{m(1 - \beta^2)^{1/2}}, \end{aligned} \quad (\text{B3})$$

$$\begin{aligned} & \ddot{\xi}_2 - b[\xi_2^{(3)} - c^{-2} \dot{\xi}_2(\ddot{\xi}_1^2 + \ddot{\xi}_2^2 - c^2 \dot{t}^2)] \\ & + b^2[\xi_2^{(4)} - 3c^{-2} \dot{\xi}_2(\ddot{\xi}_1 \xi_1^{(3)} + \ddot{\xi}_1 \xi_1^{(3)} - c^2 \dot{t} t^{(3)}) - c^{-2} \ddot{\xi}_2(\ddot{\xi}_1^2 + \ddot{\xi}_2^2 - c^2 \dot{t}^2)] \\ & = -\frac{e}{mc} \dot{\xi}_1 B + \frac{e E_2}{m(1 - \beta^2)^{1/2}}, \end{aligned} \quad (\text{B4})$$

$$\begin{aligned} & \ddot{t} - b[t^{(3)} - c^{-2} \dot{t}(\ddot{\xi}_1^2 + \ddot{\xi}_2^2 - c^2 \dot{t}^2)] \\ & + b^2[t^{(4)} - 3c^{-2} \dot{t}(\ddot{\xi}_1 \xi_1^{(3)} + \ddot{\xi}_1 \xi_1^{(3)} - c^2 \dot{t} t^{(3)}) - c^{-2} \ddot{t}(\ddot{\xi}_1^2 + \ddot{\xi}_2^2 - c^2 \dot{t}^2)] \\ & = \frac{e}{c^3} \frac{\mathbf{E} \cdot \mathbf{u}}{(1 - \beta^2)^{3/2}}. \end{aligned} \quad (\text{B5})$$

Based on the original LD equation, Zhang [33] showed that in the $(\mathbf{E}, \mathbf{B})$ field the electron with velocity βc can have a circular orbit with a constant angular velocity which satisfies

$$\beta \gamma^2 = \frac{mcE}{eB^2 b}, \quad (\text{B6})$$

$$\gamma \left(\frac{d\theta}{dt} \right) = -\frac{eB}{mc}. \quad (\text{B7})$$

However, the conditions (B6) and (B7) cannot make the electrons to keep circular orbit based on the modified LD equations. We assume the difference for an circular orbit in this case can be solved using the perturbation method with the Zhang's solution as the zeroth order. Using the same simplification notes in (26) and (27), equations (B3) to (B5) can be rewritten as

$$\dot{q} - \eta \ddot{q} + \eta \sinh q \cosh q \dot{\theta}^2 = \frac{E}{B\sqrt{\eta}}, \quad (\text{B8})$$

$$\dot{\theta} - \eta(2\dot{q}\dot{\theta} \coth q + \ddot{\theta}) + \eta^2 \dot{\theta}^3 \cosh^2 q_0 = -\frac{1}{\sqrt{\eta}}. \quad (\text{B9})$$

The first order solution is

$$\dot{\theta}_1 = \cosh^2 q_0 = \gamma_0^2, \quad (\text{B10})$$

$$\dot{q}_1 = 2 \sinh q_0 \cosh^3 q_0. \quad (\text{B11})$$

Therefore the electron whose velocity satisfies the condition (B6) and (B7) will be actually accelerated and spiral out with a slower angular momentum based on the modified LD equation. This effect can also principally considered as one method for us to have choice between LD equation and its modified form.

APPENDIX C: GENERAL ELIEZER'S EQUATION WITH HIGHER DERIVATIVES OF VELOCITY

In this appendix we take the natural unit $c = 1$ for simplicity. Following Dirac [3] and Eliezer [5], the function $\mathcal{D}^\mu$ in equation (96) should satisfy

$$v^\mu \dot{\mathcal{D}}_\mu = 0. \quad (\text{C1})$$

Dirac [3] chose the simplest form $\mathcal{D}_\mu = f v_\mu$. Eliezer [5] considered the general form

$$\mathcal{D}^\mu = P_0 v^\mu + P_1 \dot{v}^\mu + P_2 \ddot{v}^\mu + \dots + P_i v^{\mu(i)}. \quad (\text{C2})$$

We give a more general way to derive the expression of $\mathcal{D}^\mu$ which can satisfy (C1) in this appendix. Derivate the equation $v_\mu v^\mu = -1$, we have

$$C_{2i}^0 v_\mu^{(2i)} v^\mu + C_{2i}^1 v_\mu^{(2i-1)} \dot{v}^\mu + C_{2i}^2 v_\mu^{(2i-2)} \ddot{v}^\mu + \dots + \frac{i}{2} C_{2i}^i v_\mu^{(i)} v^{\mu(i)} = 0, \quad (\text{C3})$$

$$C_{2i+1}^0 v_\mu^{(2i+1)} v^\mu + C_{2i+1}^1 v_\mu^{(2i)} \dot{v}^\mu + \dots + C_{2i+1}^i v_\mu^{(i+1)} v^{\mu(i)} = 0, \quad (\text{C4})$$

where C_m^n is the notation of $\frac{m!}{n!(m-n)!}$. From equations (C3) and (C4) you can test that $\mathcal{D}_\mu$ can only contain the even terms $P_{2i} v_{2i}^{(2i)}$. Taking

$$(P_0 \dot{v}_\mu + P_2 v_\mu^{(3)} + \dots + P_{2i} v_\mu^{(2i+1)}) v^\mu = 0, \quad (\text{C5})$$

we have the equation

$$\begin{aligned} P_0 &= f - \frac{3}{2} P_2 \dot{v}_\mu \dot{v}^\mu - \dots \\ &- P_{2i} \left\{ C_{2i+1}^1 v_\mu^{(2i-1)} \dot{v}^\mu + [C_{2i+1}^2 - C_{2i+1}^1] v_\mu^{(2i-2)} \ddot{v}^\mu + \dots + \frac{[C_{2i+1}^i - C_{2i+1}^{i-1}]}{2} v_\mu^{(i)} v^{\mu(i)} \right\}, \end{aligned} \quad (\text{C6})$$

where k is the same expression as in Dirac [3], $\mathcal{D}_\mu = f v_\mu$. If we define

$$D_{2j+1}^j = C_{2j+1}^j - C_{2j-1}^{j-1} = \frac{2(i-j+1)}{j} C_{2i+1}^{j-1}, \quad (\text{C7})$$

then $\mathcal{D}_\mu$ can be written as

$$\mathcal{D}_\mu = v_\mu \mathfrak{D} + P_2 \ddot{v}_\mu + P_4 v_\mu^{(4)} + \dots + P_{2i} v_\mu^{(2i)}, \quad (\text{C8})$$

where the invariant term $\mathfrak{D}$ is

$$\begin{aligned} \mathfrak{D} &= f - \frac{3}{2} P_2 \dot{v}^2 - 5 P_4 v_\mu^{(3)} \dot{v}^\mu - \frac{5}{2} P_4 \ddot{v}^2 - P_6 \left[7 v_\mu^{(5)} \dot{v}^\mu + 14 v_\mu^{(4)} \ddot{v}^\mu + \frac{21}{2} v^{(3)^2} \right] - \dots \\ &- P_{2i} \left[C_{2i+1}^1 v_\mu^{(2i-1)} \dot{v}^\mu + D_{2i+1}^2 v_\mu^{(2i-2)} \ddot{v}^\mu + \dots + \frac{D_{2i+1}^i}{2} v_\mu^{(i)} v^{\mu(i)} \right]. \end{aligned} \quad (\text{C9})$$

Equation (C8) with (C9) is the general form of LD equation. However, this is a pure mathematical expression whose parameters P_{2i} should be further determined and explained physically. Eliezer [5] considered the case of $P_2 \neq 0$, and discussed the possible value of P_2 . Our purpose to mention this general equation here is not for a mathematica game, but to introduce a basic way to check the value of P_2 or even more parameters in (C9) physically, i.e., to calculate and measure the radiation emission which can reflect the basic motion of electrons with the parameters.

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- [1] M. Abraham and R. Becker, *Theorie der Electricitat* (Springer, Leipzig, 1933), Vol. II.
 - [2] H. A. Lorentz, 1952, *Theory of Electrons* (Dover, New York, 1952).
 - [3] P. A. M. Dirac, Proc. R. Soc. Lond. A, 167, 148 (1938)
 - [4] E. Poisson, 1999, gr-qc/9912045

- [5] C. J. Eliezer, *Mod. Phys. Rev.* 19, 147 (1945).
- [6] C. J. Eliezer, *Phys. Rev.* 71, 49 (1947).
- [7] C. J. Eliezer, *Proc. R. Soc. Lond. A*, 194, 543 (1948).
- [8] L. D. Landau and E. M. Lifshitz, *The Classical Theory of Fields* (Pergamon, London, 1962), Sec. 76.
- [9] T. C. Mo and C. H. Papas, *Phys. Rev. D* 4, 3566 (1971).
- [10] J. C. Herrera, *Phys. Rev. D* 15, 453 (1977).
- [11] F. Rohrlich, *Phys. Rev. E*, 77, 046609 (2008).
- [12] B. S. DeWitt and R. Brehme, *Ann. Phys. (N.Y.)* 9, 220 (1960).
- [13] E. Poisson, *Classical Quantum Gravity* 21, R153 (2004).
- [14] D. White, *Phys. Rev. D* 5, 1930 (1972).
- [15] A. Higuchi, *Phys. Rev. D* 66, 105004 (2002).
- [16] H. Nomura, M. Sasaki, and K. Yamamoto, *J. Cosmol. Astropart. Phys.* 11013(2006).
- [17] A. Higuchi and P. J. Walker *Phys. Rev. D* 79, 5023 (2009).
- [18] A. Higuchi and G. D. R. Martin, *Phys. Rev. D* 70, 081701(R) (2004).
- [19] A. Higuchi and G. D. R. Martin, *Phys. Rev. D* 74, 125002 (2006).
- [20] I. V. Sokolov et al., *Phys. Plasmas* 16, 093115 (2009).
- [21] I. V. Sokolov et al., physics.plasm-ph: 1003.0806
- [22] J. C. Kasher, *Phys. Rev. D* 14, 939 (1976).
- [23] E. Comay, *J. Phys. A: Math. Gen.* 29 2111 (1996).
- [24] S. G. Rajeev, *Ann. Phys.* 323, 2654C2661 (2008).
- [25] C. E. Aguiar and F. A. Baronea, *Am. J. Phys.*, 77, 4, (2009).
- [26] F. V. Hartemann, D. J. Gibson, and A. K. Kerman, *Phys. Rev. E* 72, 026502 (2005).
- [27] C. S. Shen, *Phys. Rev. Lett.* 24, 410 (1970).
- [28] B. Finkbeiner, H. Herold, T. Ertl and H. Ruder, *A&A*, 225, 479 (1989).
- [29] B. Finkbeiner and H. Herold, and H. Ruder, *A&A*, 238, 462 (1990).
- [30] R. W. Nelson and I. Wasserman, *ApJ*, 371, 265 (1991).
- [31] D. Vogt and P. S. Letelier, *Gen. Relativ. Gravit.* 35, 2261 (2003).
- [32] G. B. Rybicki and A. P. Lightman, *Radiative Processes in Astrophysics*, (Wiley-VCH, New York, 2004), Chapter 3.
- [33] Z. S. Zhang, 2004, *Electrodynamics and Special Relativity*, (Peking Univ Press, Beijing, 2004).
- [34] J. C. Herrera, *Phys. Rev. D* 7, 1567 (1973).
- [35] J. C. Herrera, *Phys. Rev. D* 10, 4297 (1974).
- [36] N. D. Lubart, *Phys. Rev. D* 9, 2717 (1974).
- [37] R. C. Duncan and C. Thompson, *ApJL*, 392, 9 (1992).